

VIROLAHTI KOIVUNIEMI, Finland

WMO: 028310

Lat: 60.527N

Lon: 27.673E

Elev: 5

StdP: 101.26

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.2	-20.6	-26.8	0.3	-24.1	-22.9	0.5	-20.5	9.1	1.4	8.4	0.8	1.1	0	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	26.8	19.5	24.9	18.8	23.1	17.5	21.1	25.1	19.7	23.6	18.5	22.0	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.2	23.5	18.2	13.1	22.0	17.0	12.2	20.5	61.0	25.1	56.5	23.5	52.4	22.2	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.4	6.6	-25.4	27.8	5.8	2.7	-29.6	29.7	-33.0	31.3	-36.3	32.8	-40.5	34.7		
			-25.6	21.8	5.8	1.4	-29.8	22.8	-33.2	23.6	-36.4	24.4	-40.6	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-6.9	-5.5	-2.5	3.0	9.5	13.5	16.6	15.4	11.0	5.0	1.7	-2.4	
	DBStd	9.04	6.95	6.33	4.44	3.17	4.18	3.24	3.22	2.90	3.13	3.57	4.15	5.55	
	HDD10.0	2438	524	433	387	210	60	7	1	2	24	158	249	383	
	HDD18.3	4924	782	666	646	459	274	149	74	100	219	414	499	641	
	CDD10.0	587	0	0	0	1	45	112	204	167	55	2	0	0	
	CDD18.3	32	0	0	0	0	1	3	21	8	0	0	0	0	
	CDH23.3	351	0	0	0	0	16	53	208	73	0	0	0	0	
	CDH26.7	52	0	0	0	0	1	6	37	9	0	0	0	0	
Wind	WSAvg	3.1	3.1	3.0	3.2	2.8	3.1	3.2	2.8	2.9	3.0	3.3	3.5	3.7	
Precipitation	PrecAvg	686	57	49	41	33	41	55	59	75	67	76	72	62	
	PrecMax	952	108	112	86	74	110	134	130	173	153	188	168	145	
	PrecMin	500	4	1	2	6	9	8	3	5	14	20	7	13	
	PrecStd	102	24	27	20	18	27	26	31	39	42	43	34	28	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	4.9	7.6	18.2	25.3	27.3	29.6	27.8	21.1	13.7	10.2	7.6	
		MCWB	3.3	2.6	3.5	10.8	16.2	18.6	21.1	20.2	16.8	11.6	9.3	7.3	
	2%	DB	3.1	3.0	5.6	14.3	22.9	24.8	27.6	25.4	19.1	12.6	8.9	6.0	
		MCWB	2.6	1.9	2.8	8.5	15.0	18.0	20.5	19.5	15.6	10.9	8.1	5.4	
	5%	DB	2.2	2.0	4.0	11.2	20.6	22.6	25.8	23.2	17.8	11.7	7.7	4.9	
		MCWB	1.9	1.5	2.0	6.8	13.6	16.1	19.9	18.1	14.8	10.4	7.1	4.3	
	10%	DB	1.1	1.3	2.7	9.3	18.3	20.6	23.7	21.4	16.6	10.7	6.8	3.8	
		MCWB	0.9	0.9	1.4	5.6	12.4	15.4	18.6	17.3	14.3	9.5	6.3	3.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.0	4.5	11.2	18.2	20.6	22.9	22.1	17.6	12.5	9.5	7.3
			MCDB	3.8	4.2	6.0	16.2	23.3	25.1	27.3	26.5	20.0	13.4	10.0	7.5
2%		WB	2.7	2.2	3.3	9.2	16.1	18.8	21.6	20.3	16.4	11.6	8.2	5.5	
		MCDB	3.0	2.8	4.7	13.3	20.5	23.3	26.1	23.7	18.3	12.2	8.7	5.9	
5%		WB	1.9	1.6	2.3	7.5	14.5	17.2	20.6	19.0	15.6	10.8	7.2	4.3	
		MCDB	2.2	2.0	3.4	10.5	19.2	21.3	24.6	22.5	17.1	11.5	7.6	4.8	
10%		WB	0.9	1.0	1.6	6.1	12.8	16.0	19.3	17.8	14.7	9.7	6.4	3.2	
		MCDB	1.1	1.3	2.5	8.5	17.5	19.6	23.0	20.5	16.1	10.6	6.8	3.7	
Mean Daily Temperature Range		5% DB	MDBR	6.0	5.8	7.3	9.1	11.2	9.9	9.9	9.8	8.3	6.3	4.0	4.5
			MCDBR	3.2	4.2	7.7	13.7	14.7	13.8	13.0	13.1	10.4	7.5	4.5	4.2
	MCWBR		3.0	3.8	5.7	8.7	8.0	7.4	7.0	7.7	7.5	6.2	4.4	4.3	
	5% WB	MCDBR	3.1	4.0	6.2	12.1	12.9	11.8	11.2	10.9	8.4	6.4	4.5	4.1	
		MCWBR	3.0	3.7	4.8	8.2	7.5	7.2	6.5	6.9	6.4	5.7	4.4	4.2	
Clear-Sky Solar Irradiance	taub	0.275	0.310	0.332	0.370	0.367	0.365	0.388	0.378	0.353	0.333	0.302	0.381		
	taud	2.122	2.150	2.200	2.284	2.408	2.457	2.412	2.433	2.488	2.383	2.198	2.591		
	Ebn at Noon	461	661	781	822	858	866	833	811	771	646	425	319		
	Edh at Noon	42	74	97	108	103	100	103	93	75	60	40	28		
All-Sky Solar Radiation	RadAvg	0.25	0.86	2.35	3.85	5.26	5.49	5.48	4.18	2.48	1.02	0.27	0.13		
	RadStd	0.06	0.18	0.29	0.41	0.57	0.36	0.49	0.43	0.28	0.15	0.05	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (45 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	-175	-189	N/A	N/A

Nomenclature: See separate page